

An Analysis of Retrieval augmented generation

Retrieval augmented generation

Retrieval augmented generation -- Part 1

Performance improves by optimizing how vector similarities are calculated. Dot products enhance similarity scoring, while approximate nearest neighbor (ANN) searches improve retrieval efficiency over K-nearest neighbors (KNN) searches. Accuracy may be improved with Late Interactions, which allow the system to compare words more precisely after retrieval. This helps refine document ranking and improve search relevance. Hybrid vector approaches may be used to combine dense vector representations with sparse one-hot vectors, taking advantage of the computational efficiency of sparse dot products over dense vector operations. Other retrieval techniques focus on improving accuracy by refining how documents are selected. Some retrieval methods combine sparse representations, such as SPLADE, with query expansion strategies to improve search accuracy and recall.

Retrieval augmented generation -- Part 2

RAG poisoning RAG systems may retrieve factually correct but misleading sources, leading to errors in interpretation. In some cases, an LLM may extract statements from a source without considering its context, resulting in an incorrect conclusion. Additionally, when faced with conflicting information, RAG models may struggle to determine which source is accurate. The worst case outcome of this limitation is that the model may combine details from multiple sources producing responses that merge outdated and updated information in a misleading manner. According to the MIT Technology Review, these issues occur because RAG systems may misinterpret the data they retrieve.

Retrieval augmented generation -- Part 3

Applications Retrieval-augmented generation is used in applications where generated responses need to be grounded in external or frequently updated information. Commonly cited use cases include search engines, question-answering systems, customer support chatbots, enterprise knowledge assistants, content generation, recommendation systems, retail and e-commerce, and industrial or manufacturing workflows. In healthcare, RAG has been studied as a way to ground large language model outputs in external medical knowledge sources, although reviews have noted continuing challenges around evaluation, ethics, and clinical reliability.

Retrieval augmented generation -- Part 4

Retrieval-augmented generation (RAG) is a technique that enables large language models (LLMs) to retrieve and incorporate new information from external data sources. With RAG, LLMs first refer to a specified set of documents, then

respond to user queries. These documents supplement information from the LLM's pre-existing training data. This allows LLMs to use domain-specific and/or updated information that is not available in the training data. For example, this helps LLM-based chatbots access internal company data or generate responses based on authoritative sources. RAG improves LLMs by incorporating information retrieval before generating responses. Unlike LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. According to Ars Technica, "RAG is a way of improving LLM performance, in essence by blending the LLM process with a web search or other document look-up process to help LLMs stick to the facts." This method helps reduce AI hallucinations, which have caused chatbots to describe policies that don't exist, or recommend nonexistent legal cases to lawyers that are looking for citations to support their arguments. RAG also reduces the need to retrain LLMs with new data, saving on computational and financial costs. Beyond efficiency gains, RAG also allows LLMs to include sources in their responses, so users can verify the cited sources. This provides greater transparency, as users can cross-check retrieved content to ensure accuracy and relevance. The term retrieval-augmented generation (RAG) was introduced in a 2020 paper that described combining a parametric language model with a non-parametric external memory accessed through retrieval at inference time.

Retrieval augmented generation -- Part 5

Hybrid search Sometimes vector database searches can miss key facts needed to answer a user's question. One way to mitigate this is to do a traditional text search, add those results to the text chunks linked to the retrieved vectors from the vector search, and feed the combined hybrid text into the language model for generation. The adoption of RAG in consumer-facing web search products has given rise to new content optimization disciplines, as practitioners have noted that content retrievability in RAG systems depends on factors like semantic structure, passage-level authority signals, and entity clarity rather than traditional search ranking signals such as backlinks.

Retrieval augmented generation -- Part 6

Pre-training the retriever using the Inverse Cloze Task (ICT), a technique that helps the model learn retrieval patterns by predicting masked text within documents. Supervised retriever optimization aligns retrieval probabilities with the generator model's likelihood distribution. This involves retrieving the top-k vectors for a given prompt, scoring the generated response's perplexity, and minimizing KL divergence between the retriever's selections and the model's likelihoods to refine

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retrieval. Reranking techniques can refine retriever performance by prioritizing the most relevant retrieved documents during training.

Retrieval augmented generation -- Part 7

By redesigning the language model with the retriever in mind, a 25-time smaller network can get comparable perplexity as its much larger counterparts. Because it is trained from scratch, this method (Retro) incurs the high cost of training runs that the original RAG scheme avoided. The hypothesis is that by giving domain knowledge during training, Retro needs less focus on the domain and can devote its smaller weight resources only to language semantics. The redesigned language model is shown here. It has been reported that Retro is not reproducible, so modifications were made to make it so. The more reproducible version is called Retro++ and includes in-context RAG.

Retrieval augmented generation -- Part 8

RAG and LLM limitations LLMs can provide incorrect information. For example, when Google first demonstrated its LLM tool "Google Bard" (later re-branded to Gemini), the LLM provided incorrect information about the James Webb Space Telescope. This error contributed to a \$100 billion decline in Google's stock value. RAG is used to prevent these errors, but it does not solve all the problems. For example, LLMs can generate misinformation even when pulling from factually correct sources if they misinterpret the context. MIT Technology Review gives the example of an AI-generated response stating, "The United States has had one Muslim president, Barack Hussein Obama." The model retrieved this from an academic book rhetorically titled Barack Hussein Obama: America's First Muslim President? The LLM did not "know" or "understand" the context of the title, generating a false statement. LLMs with RAG are programmed to prioritize new information. This technique has been called "prompt stuffing." Without prompt stuffing, the LLM's input is generated by a user; with prompt stuffing, additional relevant context is added to this input to guide the model's response. This approach provides the LLM with key information early in the prompt, encouraging it to prioritize the supplied data over pre-existing training knowledge.